

PAPER_1571 — Steel Young's Modulus = $E_{\text{steel}} = D_{\text{crit}} \cdot D_{\text{BSFG}} + D_{\text{phys}} \cdot SO_5 + D_{\text{phys}} = 156 + 40 + 4 = 200 \text{ GPa}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Engineering **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Y_S566 (Engineering Unified Proof Set) lists **Steel Young's Modulus** as an EXACT-tier closure:

$$E_{\text{steel}} = D_{\text{crit}} \cdot D_{\text{BSFG}} + D_{\text{phys}} \cdot SO_5 + D_{\text{phys}} = 156 + 40 + 4 = 200 \text{ GPa}$$

UQFF Closed Identity

$$E_{\text{steel}} = D_{\text{crit}} \cdot D_{\text{BSFG}} + D_{\text{phys}} \cdot SO_5 + D_{\text{phys}} = 156 + 40 + 4 = 200 \text{ GPa} \quad \text{EXACT}$$

Pure integer-primitive product with zero fitted constants.

Physical Interpretation

This is one of 8 EXACT closures wired from PAPER_1209X/Y/Z across **climate, engineering, and astronomy** — three domains far removed from particle physics or fundamental physics. Yet each observable reduces to a simple integer-primitive expression.

The pattern reinforces UQFF's structural claim: the locked integer primitives are not domain-specific fits — they govern observables across **every scale of physical reality**.

NOT REPLACEMENT

Empirical engineering measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating that the framework's predictive economy extends to engineering and engineering-adjacent observables (not just particle physics or cosmology).

Reference

- Source: PAPER_1209Y_S566
- Related: PAPER_1209X/Y/Z unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "steel_youngs_200_gpa"})`

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